ExpNo-13 Experiment to handle missing values and inappropriate data in a given data set

Program:

```
import numpy as np

import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")

df

# identify duplicate row

df.duplicated()

df.info()

# remove duplicate rows

df.drop_duplicates(inplace=True)

df

len(df)

# Reset the index

index = np.arange(0, len(df))

df.set_index(index, inplace=True)

index

df

# remove duplicate column

df.drop(['Age_Group.1'], axis=1, inplace=True)

df

# replace inappropriate negative values with NaN

df.loc[df['CustomerID'] < 0, 'CustomerID'] = np.nan

df.loc[df['Bill'] < 0, 'Bill'] = np.nan

df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan

df
```

```
df.loc[(df['NoOfPax'] < 1) | (df['NoOfPax'] > 20), 'NoOfPax'] = np.nan
df
df.Age_Group.unique()
df.Hotel.unique()
df['Hotel'] = df['Hotel'].replace(['Ibys'], 'Ibis')
df
df.FoodPreference.unique()
df['FoodPreference'] = df['FoodPreference'].replace(
  ['Vegetarian', 'veg'], 'Veg'
)
df['FoodPreference'] = df['FoodPreference'].replace(
  ['non-Veg'], 'Non-Veg'
)
# Fill missing values
df['EstimatedSalary'] = df['EstimatedSalary'].fillna(
  round(df['EstimatedSalary'].mean())
)
df['NoOfPax'] = df['NoOfPax'].fillna(
  round(df['NoOfPax'].median())
)
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(
  round(df['Rating(1-5)'].median())
)
df['Bill'] = df['Bill'].fillna(
  round(df['Bill'].mean())
)
df
```

Output:

<class 'pandas.DataFrame'>

RangeIndex: 16 entries, 0 to 15

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	CustomerID	16 non-null	int64
1	Age_Group	16 non-null	str
2	Age_Group.1	16 non-null	str
3	Hotel	16 non-null	str
4	FoodPreference	16 non-null	str
5	NoOfPax	15 non-null	float64
6	Bill	15 non-null	float64
7	EstimatedSalary	15 non-null	float64
8	Rating(1-5)	14 non-null	float64

dtypes: float64(4), int64(1), str(4)

memory usage: 1.3 KB

	CustomerID	Age_Group	Hotel	FoodPreference	NoOfPax	Bill	EstimatedSalary	Rating (1-5)
0	101.0	18-25	Ibis	Veg	2.0	1500.0	25000.0	4.0
1	102.0	26-35	Ibis	Veg	4.0	2500.0	35000.0	5.0
2	103.0	36-45	Taj	Non-Veg	3.0	2723.0	43692.0	3.0
3	NaN	26-35	Marriott	Veg	3.0	3200.0	43692.0	4.0

	CustomerID	Age_Group	Hotel	FoodPreference	NoOfPax	Bill	EstimatedSalary	Rating (1-5)
4	105.0	46-55	Ibis	Non-Veg	3.0	1800.0	60000.0	4.0
5	106.0	18-25	Taj	Veg	2.0	2723.0	30000.0	3.0
6	107.0	36-45	Ibis	Veg	5.0	3000.0	40000.0	5.0
7	108.0	26-35	Marriott	Non-Veg	3.0	4500.0	55000.0	4.0
8	109.0	46-55	Ibis	Veg	3.0	1200.0	28000.0	4.0
9	110.0	18-25	Taj	Non-Veg	4.0	2800.0	50000.0	2.0
10	111.0	36-45	Marriott	Veg	2.0	3500.0	65000.0	4.0
11	112.0	26-35	Ibis	Veg	6.0	4000.0	42000.0	5.0
12	113.0	18-25	Taj	Non-Veg	3.0	1700.0	32000.0	3.0
13	114.0	46-55	Ibis	Veg	2.0	2600.0	58000.0	4.0
14	115.0	36-45	Marriott	Non-Veg	4.0	3100.0	48000.0	5.0